PROBLEM 1.5

Algor Mortis • Thermal Inversion • Newton Cooling

Forensic Time of Death Estimation (Algor Mortis)

Original statement · ProblemsCh1.pdf, p. 2

PHASE 1: PROBLEM STATEMENT & HYPOTHESES

Problem statement: A body is found in a cold room ($T_A = 5^\circ\mathrm{C}$) at 3 pm, and its temperature then is $19^\circ\mathrm{C}$. An hour later its temperature has dropped to $15^\circ\mathrm{C}$. Use Newton's law of cooling to estimate the time of death, assuming that the living body temperature is $37^\circ\mathrm{C}$ and measuring time in hours.

PHASE 2: FUNDAMENTAL FORMULATION

Let $t = 0$ h correspond to 3:00 pm. Known data: $T_A = 5^\circ\mathrm{C}$, $T(0) = 19^\circ\mathrm{C}$, and $T(1) = 15^\circ\mathrm{C}$ (at 4:00 pm). We seek the negative time $t_d < 0$ when $T(t_d) = 37^\circ\mathrm{C}$.

PHASE 3: STEP-BY-STEP DERIVATION

Step 1: General Solution:

$$\frac{dT}{dt} = -k(T-5) \implies \frac{d(T-5)}{T-5} = -k\,dt \implies T(t) - 5 = [T(0)-5]e^{-kt}$$

With $T(0) = 19^\circ\mathrm{C}$, the temperature difference is $19 - 5 = 14^\circ\mathrm{C}$:

$$T(t) = 5 + 14e^{-kt} \tag{1.5.1}$$

Step 2: Determination of k:

At $t = 1$ h, $T(1) = 15^\circ\mathrm{C}$:

$$15 = 5 + 14e^{-k} \implies 10 = 14e^{-k} \implies e^{-k} = \frac{10}{14} = \frac{5}{7}$$

Inverting both sides:

$$e^{k} = \frac{7}{5} = 1.4 \implies \mathbf{k = \ln(1.4) \approx 0.336472\ h^{-1}}$$

Step 3: Calculating Time of Death t_d:

Enforce the living core body temperature $T(t_d) = 37^{\circ}\text{C}$:

$$37 = 5 + 14e^{-kt_d} \implies 32 = 14e^{-kt_d} \implies e^{-kt_d} = \frac{32}{14} = \frac{16}{7}$$

Take the natural logarithm:

$$-kt_d = \ln\left(\frac{16}{7}\right) \implies t_d = -\frac{\ln(16/7)}{k} = -\frac{\ln(16/7)}{\ln(1.4)}$$

Evaluating numerically with $\ln(16/7) \approx 0.8266786$ and $\ln(1.4) \approx 0.3364722$:

$$t_d = -\frac{0.8266786}{0.3364722}\ \text{h} \approx \mathbf{-2.4569}\ \text{hours}$$

Step 4: Clock Time Conversion:

Multiply the fractional part by 60 minutes:

$$2.4569\ \text{hours} = 2\ \text{h} + (0.4569 \times 60)\ \text{min} = 2\ \text{h}\ 27.41\ \text{min} \approx 2\ \text{hours and}\ 27\ \text{minutes}$$

Subtracting 2 hours and 27 minutes from 3:00 pm (15:00):

$$15:00 - 2\ \text{h}\ 27\ \text{min} = \mathbf{12}:\mathbf{33}\ \text{pm}$$

PHASE 4: FINAL RESULT & ANALYSIS

BOXED RESULT

$$t_d \approx -2.457 \text{ hours} \implies \text{Time of Death} \approx \mathbf{12} : \mathbf{33} \text{ pm}$$

Forensic Consistency: Algor mortis cooling is monotonic and deterministic under steady ambient conditions.

Study notes written from the official course material (ProblemsCh1.pdf; Robinson (2004).pdf). Online version: subjects/advanced-maths/problemas/topic-1-problems.html